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A post-hurricane building debris estimation workflow enabled by uncertainty-aware AI and crowdsourcing

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ABSTRACT

Climate disasters often result in large amounts of debris that need to be cleaned up in the event's aftermath. Effective post-disaster debris management poses unique challenges due to limited resources, high expenses, and the nonhomogeneous and hazardous debris content. Current practices of debris cleanup rely on time-consuming and error-prone field-based estimation, which impedes immediate response and accurate analysis. While new approaches such as artificial intelligence (AI) and crowdsourcing have shown promise in various domains, their potential for debris estimation remains underexplored. This study leverages a human-AI teaming workflow that estimates post-disaster debris volume and composition. The workflow begins with the utilization of drones to capture aerial views for efficient disaster infrastructure reconnaissance. Subsequently, an AI model and a crowdsourcing module are calibrated and work together to rapidly and reliably detect damaged buildings and classify the levels of damage based on aerial imagery. By establishing a connection between building damage states and the resulting generated debris, these damage level predictions serve as a foundation for the efficient and accurate estimation of debris. Overall, the proposed approach uses drones for rapid data collection and AI to automate damage detection and classification, which reduces the time required for debris estimation from days to a few hours as well as minimizes predictive uncertainty with crowdsourcing by up to 40%. A case study on Hurricane Laura in 2020 was conducted to validate the proposed approach. Findings of this study lead to predicting debris volume and composition with minimal errors, particularly for partially damaged buildings.

1. Introduction

1.1. Background

Natural hazards have become increasingly frequent and intense in recent years, resulting in severe environmental, social, and economic consequences [1,2]. Hurricanes often leave behind a vast trail of destruction, including a substantial volume of debris and waste [3]. While recognizing the profound influence of climate-induced disasters on debris generation and the ensuing cleanup efforts is crucial, the cleanup and management of this debris in the disaster's aftermath pose unique challenges, including high cost, limited resources, and the extensive geographic distribution and diverse composition of debris materials [4].

In the United States, for example, the aftermath of Hurricane Harvey in 2017 and Hurricane Ike in 2008 witnessed massive amounts of debris accumulating, resulting in substantial cleanup and removal costs. The cleanup costs in Texas alone amounted to

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$44 million after Hurricane Harvey [5] and a staggering $750 million after Hurricane Ike [6]. In fact, cleanup expenses after Hurricane Ike accounted for a significant portion, approximately 35 %, of the total public funding supported by the Federal Emergency Management Agency (FEMA) [6]. Additionally, the town of Lahaina in Hawaii has faced extensive destruction due to the 2023 Maui wildfires. The recovery and reconstruction process for this region was anticipated to extend over multiple years and necessitate a substantial financial commitment amounting to billions of dollars, as indicated by the local government of Hawaii. These statistics highlight the enormous resource and financial burden associated with post-disaster recovery and debris management.

Post-disaster debris management is also critical to the overall efficacy of disaster response and recovery efforts. Improper debris and waste management can lead to adverse environmental impacts and public health issues, further exacerbating the challenges faced during recovery [7]. For example, improper debris management can increase the risk of vector-borne diseases due to the proliferation of disease-carrying vectors like mosquitoes [8]. According to the Centers for Disease Control and Prevention (CDC), improper waste management practices that expose individuals to hazardous materials can also lead to respiratory problems and other health issues. Additionally, previous research has indicated that inadequate debris management can result in delays and complications in restoring critical infrastructure, preventing the recovery and reconstruction efforts in affected communities [1,9].

An important element in attaining effective post-disaster debris management is the precision of post-disaster debris estimation [10–12]. FEMA recommended multiplying the building's length, width, and height by a factor of 0.33 to estimate the debris quantity. Note that the 0.33 factor is to calibrate the existence of airspace within the structure [4,13]. The current practice of debris estimation involves deploying human teams to assess and estimate the amount of debris in an affected area. This can be resource-intensive in terms of personnel, time, and transportation. Manual surveys require a significant amount of human labor to visually inspect and categorize debris, often using standardized guidelines or checklists [4,14]. Furthermore, the estimation precision plays an important role in guiding the decisions that concern follow-up disaster response and recovery efforts [11,15]. For instance, Debris Task Force Leader (DTFL) of FEMA relies heavily on debris estimates when making crucial disaster-related decisions. These decisions encompass aspects such as staffing requirements, essential technical proficiency, and the overall organizational framework [4]. Debris estimation enables authorities, e.g., local government and FEMA, to allocate resources effectively and prioritize areas requiring immediate attention [16].

Despite the importance of debris estimation, the existing approach to field damage investigation and debris estimation comes with several limitations, including heavily relying on labor-intensive methods and personal judgment [9,17–19] for surveying affected regions, and gathering data concerning debris volume and composition [20]. The current practice, therefore, poses significant challenges, particularly in the aftermath of large-scale disasters, as identifying accessible and hazard-free areas for first responders may not be trivial. The time needed for manual field assessment can further impede and/or delay the cleanup operations and hinder prompt response endeavors [4]. Additionally, human estimation is inherently prone to errors, which can lead to inaccurate estimation of debris composition. Consequently, this can result in inadequate waste collection, separation, and recycling, thus compromising the accuracy of damage analysis. Ultimately, inaccurate estimation can lead to the misallocation of resources and prolonged recovery timelines. Poor estimation of debris can further hamper efforts to identify appropriate waste management strategies, including recycling and disposal methods [21,22]. Therefore, there is an urgent need for a more efficient and accurate approach for post-disaster debris estimation.

1.2. Opportunities for the use of AI, crowdsourcing, and human-AI teaming

Advancements in remote sensing, AI, and crowdsourcing hold promise for addressing the challenges associated with aerial debris detection and estimation in disaster response to enhance situational awareness [20,23,24]. Remote sensing data obtained from unmanned aerial vehicle (UAV) and satellites can provide a comprehensive and timely view of disaster-affected areas [24–30]. AI techniques, such as machine learning and computer vision, have enabled the automated analysis of remote sensing data, facilitating rapid infrastructure damage assessment [23,31]. Crowdsourcing platforms engage the public in data collection and analysis, harnessing collective intelligence to gather information in real-time for improving disaster reconnaissance practice [32–34]. For example, crowdsourcing has been effectively applied in tasks such as mapping crisis areas and annotating large datasets in the literature, demonstrating its potential to harness collective intelligence for reliable data collection and analysis [31,33,35,36].

Recently, the integration of AI technologies with human expertise has shown promising results in decision-making and problem-solving tasks across various fields [35,37–42]. Combining the computational power and data analysis capabilities of AI systems with the nuanced judgment and contextual understanding of humans leads to more accurate and reliable outcomes [33,43]. Applications of human-AI teaming span diverse domains, including healthcare, finance, cybersecurity, and disaster response. For example, Sowa et al. [44] demonstrated that human-AI collaboration in business management increased productivity through effective virtual assistance. Patel et al. [45] highlighted the benefits of human-AI teaming in medical diagnosis, improving diagnostic accuracy and decision-making. Nguyen et al. [46] proposed a human-AI teaming framework for fact-checking, enhancing accuracy and reliability through integrating human knowledge and communicating AI prediction uncertainty. Nevertheless, the challenges and limitations of these studies include addressing bias and ethical considerations, designing effective human-AI interfaces, and mitigating potential job displacement. Moreover, trust, communication, and shared understanding between humans and AI are crucial for successful collaboration, with explainability and interpretability playing important roles [42,47–49]. Current research shows an upward trend of the application of adaptive AI systems, collaborative decision-making processes, and interdisciplinary studies to deepen our understanding and improve the effectiveness of human-AI teaming. Overall, human-AI teams have the potential to revolutionize problem-solving and decision-making processes by harnessing the strengths of both human and AI systems.

Despite these advancements, the integration of human-AI teaming into post-disaster debris estimation is still relatively limited. Review papers by Sun et al. [50] and Iqbal et al. [51] have extensively discussed technological breakthroughs in the field of disaster management. These papers have identified various domains where technologies have made significant contributions, highlighting areas where their application has been limited. Both studies emphasize that disaster debris detection and estimation as one such domain where computational technologies (e.g., AI, UAVs, and crowdsourcing platforms) have been relatively underutilized.

1.3. Scope and contributions

This study aims to design and evaluate a workflow that combines AI and crowdsourcing to rapidly and accurately estimate disaster debris volume and composition. Building on the authors' prior work on post-disaster assessments using AI, UAV imagery, and crowdsourcing [33], the research presented in this paper introduces an automated debris estimation process that employs a previously developed and validated human-AI model [33], to predict damage levels as proxies for debris quantities, in accordance with FEMA's description of damage levels [14]. The central hypothesis of this research is that the predicted damage state will result in precise debris estimates since the damage definitions are closely aligned with the criteria outlined in FEMA HAZUS-MH manual [14]. This hypothesis is informed by the fact that hurricanes typically cause a change in the spatial distribution of debris, while other natural hazards (e.g., wildfires) may lead to both a change in distribution and a decrease in debris volume. This specific feature suggests the potential of developing a more accurate post-hurricane debris estimation approach by analyzing the relationship between the extent of building damage and the amount of debris generated. A case study from the 2020 Hurricane Laura will serve to test this hypothesis and demonstrate the effectiveness of the designed methodology. The presented research offers a significant advancement in automated post-disaster debris estimation by utilizing a human-AI collaborative system that merges a convolutional neural network (CNN), Bayesian inference, and crowdsourcing for post-disaster debris estimation. The CNN interprets visual sensing data, crowdsourcing enhances system's reliability by incorporating diverse perspectives and expertise from the public, while the Bayesian inference technique iteratively refine estimates, enhancing reliability and reducing the uncertainty prevalent in AI assessments. This integration aims to provide dependable and rapid debris estimation on a large scale [33,36]. The broader implications of our findings include improved disaster preparedness and response strategies, particularly in resource-constrained settings and limited accessibility. Accurate debris estimation assists in planning and optimizing resource allocation, ensuring timely and effective cleanup operations, and reducing the overall impact on affected communities. The manuscript is structured as follows: Section 2 introduces our approach workflow using AI, UAVs, and crowdsourcing in post-disaster debris estimation. Section 3 presents the implementation of the workflow, including the case study of Hurricane Laura, the AI training process, and the integration of crowdsourcing for validation. Sections 4 and 5 shows the results and discuss the performance, practical implications for disaster response, limitations of the current study, and potential future research directions.

2. Approach

A novel mapping approach from damage level to debris estimation is proposed in this research. To further automate and improve the quality of the damage level analysis by investigators, a human-AI teaming workflow is utilized with visual data of disaster-affected buildings. The workflow comprises two main processes as depicted in Fig. 1. The first process is an uncertainty-aware AI that performs visual building detection and damage classification based on the FEMA rating system [14]. Its primary purpose is to infer the levels of damage to buildings based on UAV imagery while also quantifying the associated uncertainty. The next step crowdsources opinions or knowledge from a group of humans to help judge and assess building damage where participants' reliability is measured with Bayesian inference and used to make sure their feedback is reliable. This process is designed to reduce the model-based uncertainty associated the AI-assisted damage assessment. Finally, with these combined predictions from humans and AI, the estimation of debris volume and composition is mapped and achieved based on building portfolio data and comprehensive damage criteria outlined in the FEMA manual, as shown in Fig. 2.

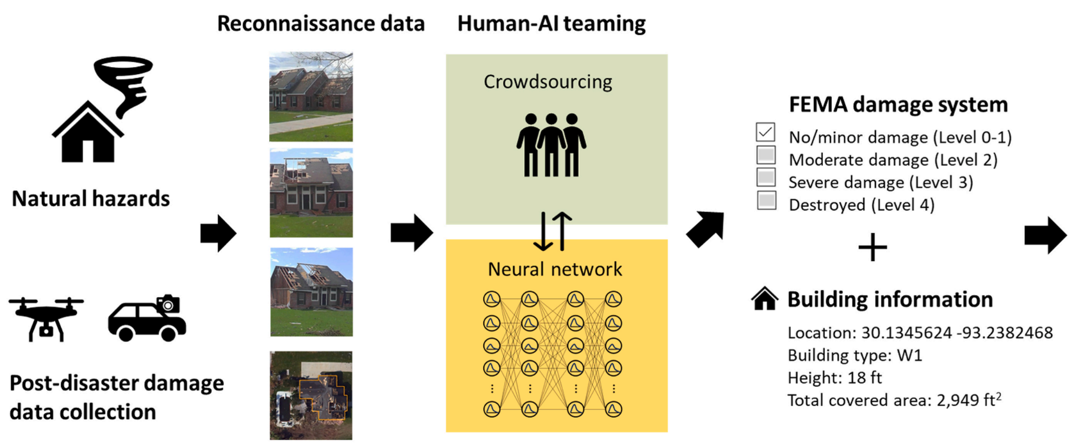

Fig. 1. Schematic visualization of the workflow for automated debris estimation.

Damage State	*Quantitative Damage Description*	*Roof Cover Failure*	*Window Door Failures*	*Roof Deck*	*Missile Impacts on Walls*	*Roof Structure Failure*	*Wall Structure*
0	**No Damage or Very Minor Damage** Little or no visible damage from the outside. No broken windows, or failed roof deck. Minimal loss of roof over, with no or very limited water penetration.	<2%	No	No	No	No	No
1	**Minor Damage** Maximum of one broken window, door or garage door. Moderate *roof* cover loss that can be covered to prevent additional water entering the building. Marks or dents on walls requiring painting or patching for repair.	>2% and <15%	One window, door failure	No	<5 impacts	No	No
2	**Moderate Damage** Major roof cover damage, moderate window breakage. Minor roof sheathing failure. Some resulting damage to interior of building from water	>15% and <50%	>1% and the larger of 20% and 3	1 to 3 panels	5 to 10 impacts	No	No
3	**Severe Damage** Major window damage or roof sheathing loss. Major roof cover loss. Extensive damage to interior from water.	>50%	> the larger of 20% & 3 and <50%	>20 impacts	>20 impacts	No	No
4	**Destruction** Complete roof failure and/or, failure of wall frame. Loss of more than 50% of roof sheathing.	>50%	>50%	>25%	>20 impacts	Yes	Yes

Fig. 2. Guidelines in FEMA HAZUS-MH hurricane module for building damage state estimation [14].

This section will begin with introducing a mapping approach between damage level and debris estimation in *section 2.1*. Following this, we introduce an AI for automated damage assessment methodology and the crowdsourcing approach in *section 2.2*. Next, *section 2.3* provides an overview of the crowdsourcing methodology.

2.1. Mapping from damage level to debris estimation

Damage level predictions generated by the human-AI teaming approach are leveraged as a substitute for achieving quick, safe, and accurate estimates of the volume and composition of structural debris. The idea is that since the damage state definitions produced by humans and AI align with the FEMA manual, which provides detailed quantitative values for wind-induced damages (as shown in Fig. 2), the resulting estimation of debris is anticipated to be accurate and reflective of the actual damaged conditions. By leveraging damage level predictions as a proxy, the estimation process can be expedited, ensuring efficient and timely assessment of the debris situation in post-disaster scenarios. This approach offers several advantages, such as reducing the need for manual and potentially hazardous assessment tasks, enabling a faster response in debris removal efforts, and providing valuable information for resource allocation and decision-making processes. The alignment between AI outcomes and the FEMA manual strengthens the confidence in the accuracy and reliability of the debris estimation, making it a valuable tool for disaster management and recovery operations.

This debris estimation workflow, as presented in Fig. 1 encompasses both volume estimation and composition estimation. It enables predicting the volumes and types of debris generated from buildings, such as roof and wall debris, as well as the volumes of post-disaster debris, utilizing the damage level descriptors provided by FEMA. Table 1 presents a proposed relationship between damage

Table 1
Connection between damage levels and debris estimation for roof and wall components.

Damage Level	Roof Cover (RC)	Roof Deck (RD)	Roof Structure (RS)	Wall Structure (WS)	Roof Debris (RDB)	Wall Debris (WD)
0	A = RC × (0%–2%)	B = RD × 0 %	C = RS × 0 %	D = WS × 0 %	RDB = A + B + C	WD = D
1	A = RC × (2%–15 %)					
2	A = RC × (15%–50 %)	B = RD × 1 %				
3	A = RC × (50%–100 %)	B = RD × (1%–25 %)				
4		B = RC × (25%–100 %)	C = RS × 100 %	D = WS × 100 %		

assessment results and debris estimation for Roof and Wall Components. Note that this proposed connection aims to enable debris estimation involves predicting both the volumes and types of debris generated from buildings. The method focuses particularly on estimating the debris generated by roofs and walls, which are known to sustain extensive damage under strong winds and can contribute to a significant portion of the total debris volume resulting from wind-induced damage to infrastructures [52].

According to the damage level descriptions in Fig. 2, the authors categorize the roofing component into three parts, as shown in Table 1: roof cover (RC), roof deck (RD), and roof structure (RS). The resulting debris from each component, referred to as RC, RD, and RS, is combined to determine the overall roof debris (RDB). For walls, the wall debris (WD) is solely determined by the debris from the wall structure (WS). The debris volumes contributed by RC, RS, and RD are denoted by A, B, and C, respectively. It is important to note that the FEMA rating system (Fig. 2) provides a range of damage assessments, and certain descriptions in Table 1 incorporate upper and lower bounds for volume estimation to accommodate prediction uncertainty.

For example, Table 1 illustrates that in the case of minor damage (i.e., damage level 1), the upper bound of roof cover debris is estimated as 15 % of the roof cover total volume. Similarly, when the damage severity reaches level 4, all roof components and wall structures (i.e., 100 % of roof covers, decks, structures, and wall structures) are completely destroyed. Note that, in this method, the focus majorly revolves around predicting the debris generated by roofs and walls after hurricane events. This is based on literature indicating that roofs and walls tend to sustain more extensive damage under strong winds [53,54]. Merely these two structural components alone can explain the majority of the total debris volume generated by wind-induced damage to infrastructure [52].

Due to the limited availability of debris data from real-life incidents, the authors conducted an experiment using a scaled model of a building (14 cm × 12 cm x 22 cm) [55], shown in Fig. 3. To simulate various damage scenarios, one of the authors inflicted systematic damage to the wood building model based on FEMA's provided descriptions as illustrated in Fig. 2. The measured debris volumes corresponding to these damage states served as the ground truth for evaluating the accuracy of our estimation method. The findings indicated that this debris estimation approach is reliable compared to existing methods, which typically demonstrate an estimation error of approximately 50 % [56]. Note that, although our pioneer experiment focuses on a small-scale wood building, the approach improves the automated visual debris estimation.

2.2. Artificial intelligence for automated damage assessment

In this workflow in Fig. 1, the AI module is designed to assess building damage while considering uncertainty, allowing for effective communication of the module's reliability by extracting uncertainty information from its predictions. To accomplish this, a Bayesian neural network is employed to incorporate probabilistic predictions of building damage severity. Uncertainty quantification in neural networks is performed using variational Bayesian inference with Monte Carlo (MC) dropout sampling [57] where the fundamental theory of variational Bayesian inference is referred to Ref. [33]. Note that neural networks traditionally use deterministic weights, but to better understand the uncertainty in their predictions, these weights as variables with a range of possibilities (i.e., probability distributions). Instead of calculating these distributions directly, which is complex, a variational Bayesian inference proposed by Graves (2011) [58] is used to approximate them by comparing two probability distributions. During model training and testing, the dropout mechanism during the inference process is activated. The model's output, represented by softmax probabilities associated with each damage severity class, is treated as random variables characterized by their probability density functions [59].

While the development of AI for damage assessment is a significant advancement in creating an efficient and scalable automated tool, there is a major challenge in utilizing these models reliably in practice. Most AI models are based on a single image, which may provide some indication but not necessarily a comprehensive description of the building's actual damage state. Classification outcomes corresponding to multiple views of the same building were shown to differ due to factors such as directional damage, partial visibility, and image quality [60]. The use of more comprehensive visual data, including multiple ground and aerial views of damaged buildings, holds the potential to improve the quantification of damage levels. Employing multiple views also aligns more closely with human assessment of the extent of building damage [60]. To this end, the damage assessment AI is extended to enable a more descriptive automated post-disaster assessment of damaged buildings through the utilization of multiple-view imagery. We, in particular, propose the use of a Multi-View Convolution Neural Network (MV-CNN) architecture, which combines information from different views (i.e., multiple inputs) of a damaged building, resulting in a 3-D aggregation of 2-D damage features from each view. The advan-

Fig. 3. Wooden house model for a small-scale experiment.

tages of incorporating multiple views for accurate and reliable classification of post-disaster building damage are explored. The developed model demonstrates promising accuracy in predicting the precise damage states of buildings.

2.3. Crowdsourcing human knowledge for reducing uncertainty

The crowdsourcing module is designed to leverage crowd knowledge to reduce the uncertainty generated by AI for reliable building damage assessment. The summarized schematic diagram in Fig. 4 illustrates the crowdsourcing methodology. This probabilistic crowdsourcing approach enables precise predictions of building damage by considering each labeling task's difficulty and individual labelers' capabilities based on the crowd's input for damage evaluation [34,36]. The key idea of this crowdsourcing method is to harness the potential of citizen participants in assigning the damage state to each of the considered buildings with minimal instruction. However, incorporating crowdsourcing and participatory data presents challenges in terms of data quality due to the unreliability of participants and the subjectivity of tasks. To address this, the approach employs a logistic function to model the probability of correct labels, taking into account the ability of each crowd annotator and the difficulty of each building damage classification task, quantified by two random variables.

To infer the damage level of buildings based on human annotations, the module applies maximum posterior (MAP) estimation and Bayesian inference [33,60]. This is achieved by using the Expectation-Maximization (EM) algorithm, an iterative stochastic algorithm [61]. For a more comprehensive understanding and detailed mathematical formulations and descriptions, readers are encouraged to refer to Refs. [33,36,60].

2.4. Calibration process for the AI model and crowdsourcing module

The AI model and crowdsourcing module calibration process involves several key steps referred to our previous work [33]. For Deep-PDA, we utilize images from hurricane-affected buildings, collected from events such as Hurricane Harvey and Hurricane Laura, and label them according to FEMA-defined damage states. The AI model is suggested to be trained using the square of the Earth Mover's Distance (EMD^2) loss function and employs a dropout to regularize the model and facilitate Bayesian uncertainty quantification. For crowdsourcing module, surveys are designed with micro-tasks for participants, distributed via crowdsourcing platforms (e.g., Amazon Mechanical Turk). Participants are provided with detailed guidelines and pictorial representations to ensure understanding. Bayesian updating and the Expectation-Maximization algorithm are used to estimate participant reliability and task difficulty, integrating AI predictions as prior knowledge.

3. Case study

The case study aims to validate the proposed debris estimation approach through an examination of real residential buildings impacted by 2020 Hurricane Laura. This section first introduces the data used in the study, including its scope and descriptions. Note that the data is derived from actual hurricane damage information collected on-site during major hurricane events. Following that, the Bayesian neural network architecture and details about its training process are introduced. Next, the discussion of the setup and calibration of the crowdsourcing module is presented. Finally, the Results and Discussion section will summarize the main findings, emphasize the significant contributions of the study, and propose potential future directions. Through empirical validation, this section demonstrates the efficacy of this approach and its potential to revolutionize the way debris estimation is performed in the aftermath of natural hazards, leading to improved response strategies and more efficient allocation of resources.

3.1. Data description

Building characteristic data, visual imagery of damaged buildings, and corresponding damage information collected in the aftermath of hurricanes are used to validate the proposed debris estimation method, employing a human-AI teaming framework. Building information is sourced, in part, from the Microsoft building footprint dataset [62]. Imagery data of affected buildings were collected by the expert team from the Structural Extreme Event Reconnaissance (StEER) network after Hurricane Laura [63]. Table 2 provides

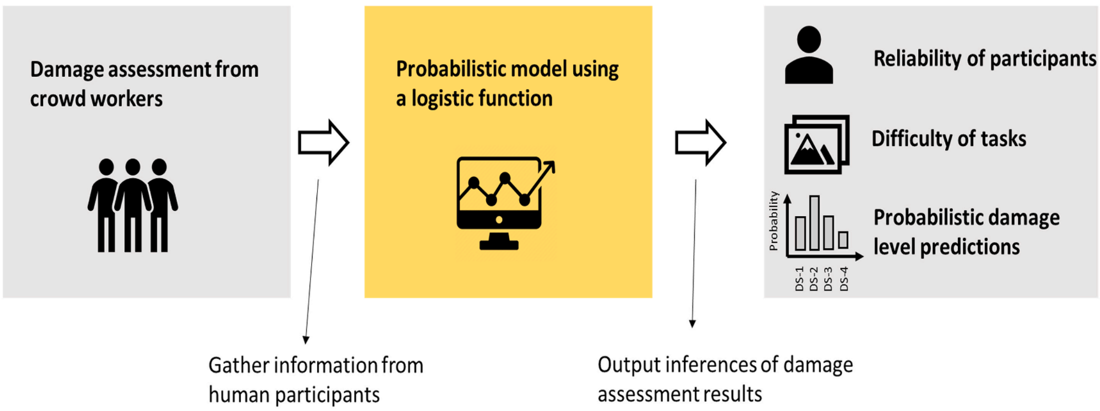

Fig. 4. The schematic representation of the crowdsourcing approach.

Table 2
Number of buildings in our database after Hurricane Laura in 2020.

	Damage Level 0	Damage Level 1	Damage Level 2	Damage Level 3	Damage Level 4	Sum
Building samples	8	35	44	4	4	95

an overview of the number of buildings in our database. The imagery includes street views capturing damaged buildings' side views, as well as top views from satellites imagery in the post-hurricane period. Expert damage inspectors conducted on-site door-to-door inspections to collect this data, which includes expert-assigned damage states and other observations that serve as ground truth for calibrating and validating the proposed methodology. Note that Hurricane Laura, a powerful category 4 storm, made landfall in Louisiana on August 27, 2020. Due to the Coronavirus pandemic that year, the expert conducted damage evaluation remotely based on visual images collected after the disaster. The need for off-site assessment and evaluation further emphasizes the importance and necessity of a reliable automated solution of damage assessment. The testing data selected for this case study consists of 95 buildings located in Lake Charles, Louisiana, USA, as indicated in Table 2.

Fig. 5 presents an example imagery corresponding to a representative building used in the dataset, providing insights into building attributes, damage severities, and damage information. It should be noted that multiple views (usually one top view and three side views in our dataset) of each building provide a comprehensive depiction of the visual damage incurred, contributing to an accurate and reliable damage assessment as introduced in *section 2.2*.

3.2. Artificial intelligence calibration

This study employs an MV-CNN architecture, visualized in Fig. 6. The MV-CNN model consists of two primary components: a multi-branch CNN network and a fully connected network in Fig. 6. The multi-branch CNN is used to extract crucial features from multiple input images and utilizes MobileNet [64] as its backbone. Particularly, a three-branch architecture is utilized in this study, incorporating one top view and two side view images. The authors demonstrated that utilizing one top view with two side views is sufficient to achieve promising performance in damage assessment [60].

The model incorporates a concatenation layer to merge the extracted information from the various CNN branches into a single layer. Following this, a fully connected neural network is employed to estimate the damage state based on the integrated feature information obtained from the multi-branch CNN. The MV-CNN generates final predictions using softmax values, which represent damage states ranging from zero to four, aligning with the FEMA rating system for residential buildings [14]. Specifically, the entire MV-CNN network is trained as an end-to-end model, enabling seamless integration of the different components for more robust predictions. To further enhance the model's predictive capabilities and generate probabilistic predictions, the Bayesian theory discussed in the methodology section is leveraged, implementing a dropout layer in every layer of the MV-CNN. These dropout layers contribute to model regularization and aid in obtaining more reliable and uncertainty-aware predictions [65].

The MV-CNN model, in particular, was initially pre-trained on the ImageNet dataset [66] and then further fine-tuned on the MV-HarveyNET dataset [67] specifically for damage assessment purposes. A dropout rate of 0.01 was selected based on [59]. During the training process, the model utilized the squared earth mover's distance (EMD^2) loss function, which can effectively account for the ordinal nature of damage level classes [68,69].

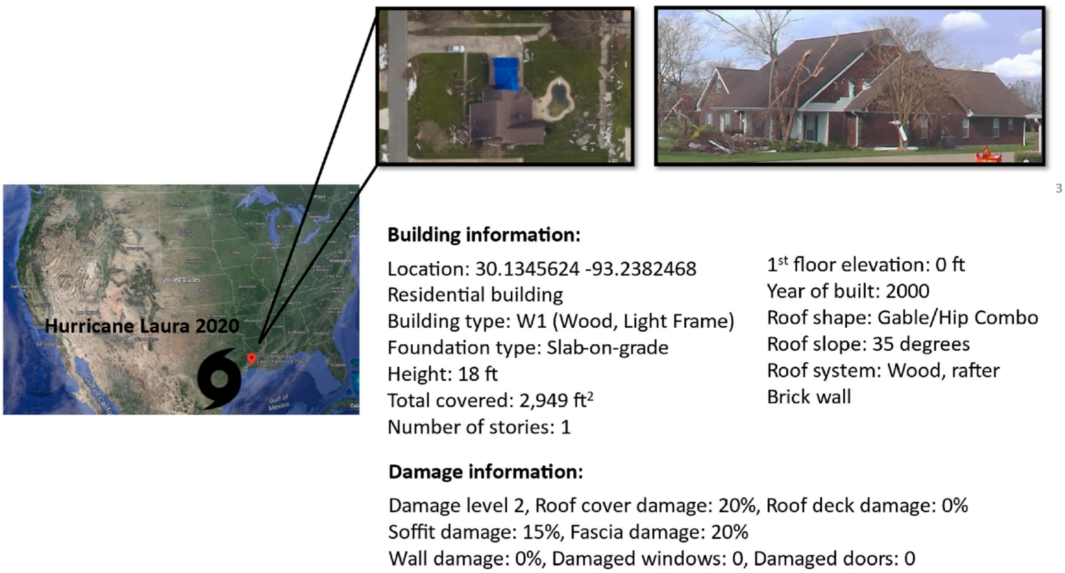

Fig. 5. Example of portfolio data available for one representative building.

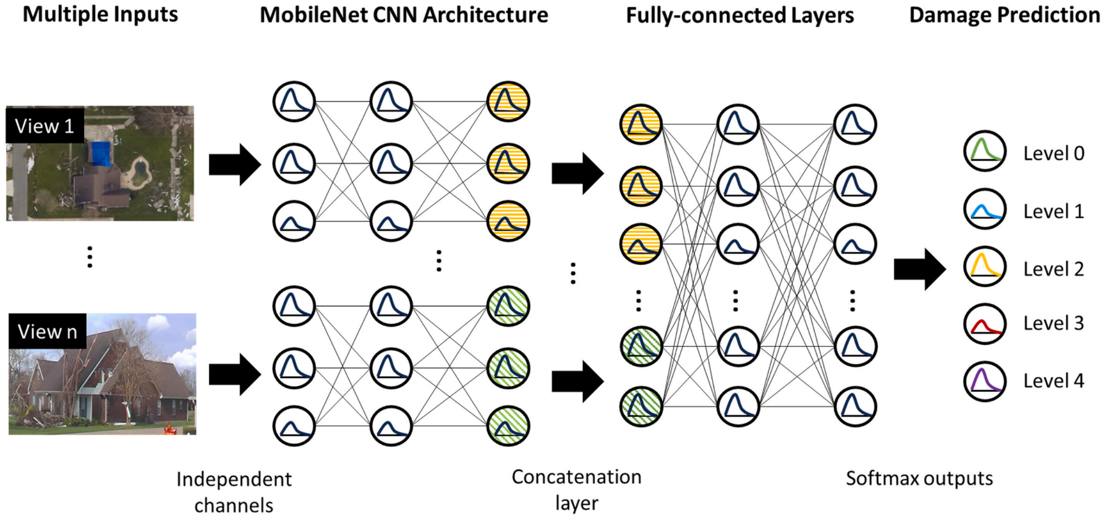

Fig. 6. The MV-CNN architecture with multiple independent input branches.

For fine-tuning the MV-CNN model, the Adam optimization algorithm [70] was used over 100 epochs, using a 10^{-4} learning rate and a 32-batch size. Data augmentation techniques were applied to ensure the robustness, involving random scaling up or down by ± 20 % and horizontal flipping of half the images. Note that these hyperparameters were not arbitrarily chosen but were empirically determined based on previous research. A sensitivity analysis was conducted to identify the most suitable hyperparameters for fine-tuning the MV-CNN model. Overall, the fine-tuning process of the MV-CNN model took approximately one and a half hours, leveraging a computing resource offered by Texas A&M University High-Performance Research Computing (HPRC) equipped with an Intel Xeon E5-2680 v4 2.40 GHz 14-core CPU, 128 GB RAM, and an NVIDIA K80 GPU with 12 GB memory.

3.3. Participatory damage assessment with crowdsourcing

3.3.1. Survey design

The task of assessing building damage based on visual characteristics is inherently subjective and requires specific expertise. However, relying solely on human experts for this assessment presents limitations such as limited availability, high costs, difficulty in accessing affected areas, and delays in rapid assessment. To address these challenges and involve non-expert citizens in the process, the authors developed a micro-tasking strategy using surveys designed to be more accessible to non-experts, while improving the accuracy and reliability of the collected data.

To create these surveys, the Qualtrics platform is utilized and distributed to crowd participants through the online crowdsourcing platform, Amazon Mechanical Turk (M-Turk). Instead of directly asking participants to assign a damage rating to a building, which can be challenging due to its subjective nature, this survey breaks down the task into simpler observational questions about specific building components. These questions align with FEMA guidelines for determining the building damage severity and guide participants in evaluating each component to make an informed decision about the overall damage state of the building.

It is worth noting that participants are provided with a sequence of micro-tasks to assist them in determining the overall damage state. Fig. 7 demonstrates the questionnaire posed to the crowd-workers to deduce the damage-state labels of the buildings. To ensure effective participation, basic guidelines are provided to respondents, including the FEMA rating system, to familiarize them with the terminology. Additionally, visual representations of typical scenarios are included to facilitate understanding and engagement.

3.3.2. Crowdsourcing setting and participant allocation

This work recruited a total of 472 participants from Amazon Mechanical Turk (M-Turk). Each participant was tasked with labeling approximately 10–15 buildings, as outlined in the Survey design for the crowdsourcing section (*section 2.3*). Note that incomplete participant responses were excluded from the analysis to ensure data integrity. In this case study, rather than involving all participants, a smaller subset of around 65 participants was allocated to assess each building. This approach was implemented to optimize efficiency and cost-effectiveness by distributing tasks among a subset of participants. Each participant was assigned a specific subset of the buildings, allowing for more focused and reliable responses to the assigned tasks.

By utilizing crowdsourcing through M-Turk, our crowdsourcing methodology considers the collective intelligence and diverse perspectives of a larger pool of participants. It ensures a comprehensive evaluation of the damage to the buildings, leveraging the varied expertise and backgrounds of the participants. Moreover, involving multiple participants for each building introduces redundancy and enables comparison among individual assessments, helping to identify potential inconsistencies or biases.

1.	Is the building structure completely destroyed (on the ground)?			
	Yes	No	Not Sure	--
2.	How much is the wall structure damaged?			
	Completely	Partially	Intact	--
3.	How much is the roof structure damaged?			
	Completely	Partially	Intact	--
4.	How much is the wall-cladding damaged?			
	Damaged and exposes the interior structure	Damaged but does not expose the interior	No/ very minor damage	--
5.	What is the condition of the roof-deck?			
	More than 25% of the roof-area damaged	Less than 25% of the roo f area damaged	1-3 panels damaged	No/Minor damage
6.	What is the condition of the roof-cover?			
	More than 50% roof-cover loss.	15-50% roof cover loss	Can be covered; 2-15% roof cover loss	No/Minor damage (less than 2%)
7.	How many windows and doors are broken?			
	More than 50% windows and doors broken	20-50%	less than 25%	Max. of 1 window, door or garage door broken

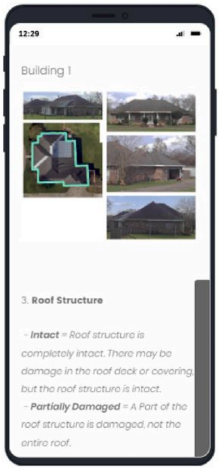

Fig. 7. The questionnaire designed for participatory damage assessment via crowdsourcing.

4. Results

In this study, a prediction of the damage level is considered acceptable if it deviates by no more than one damage state from the actual (ground-truth) value. The human-AI approach achieves an accuracy of 85 % when considering the ± 1 class interval from the ground-truth, marking an approximately 65 % improvement compared to AI predictions without human inputs with only 53 % accuracy. The collaboration between humans and AI effectively enhances the precision of building damage assessments. It is worth noting that, in a similar scenario, human experts' performance is reported to be around 90 % accurate [71], indicating that human experts' judgments are also subjective and inconsistent. The decision to report accuracy incorporating the ± 1 class deviation aligns with previous findings, which show that even experts may have differing opinions when describing building damage levels after hurricane events [68]. Although incorporating a high standard for the model to achieve "perfect" prediction is an idea, it is still worthwhile to report the model's capability to predict outcomes that closely approximate the ground truth. This approach ensures a practical evaluation of the human-AI approach's performance and its applicability in real-world scenarios where variations and uncertainties in damage assessment are expected. Overall, our approach uses drones for rapid data collection and AI to automate damage detection and classification, reducing the time required for damage and debris estimation from days to a few hours and minimizing predictive uncertainty through crowdsourcing by up to 40 %.

Next, Table 1 is utilized to establish a connection between damage assessment predictions and the estimation of debris volume and composition. As an example of the building in Fig. 5, the human-AI teaming approach can accurately predict the damage level based on FEMA guidelines. The ground-truth of this particular building is damage level 2, moderate damage. For this building in the aftermath of Hurricane Laura, RC, RD, and RS, are estimated to be approximately 78, 122, 201 cubic yards (CY), respectively, based on the information in Fig. 5. In this case of moderate to a building, the lower bound of roof cover debris (A) is estimated as 15 % of RC (11.7 CY), while A is estimated to be at most 50 % of RC (39 CY). The roof deck (RD) is estimated 1 % damaged (1.22 CY). In this scenario, the roof structure (RS) does not contribute (0 CY) to the total debris volume since it remains intact. However, in some extreme cases of completely destroyed buildings (damage state 4), 100 % of RS and wall structure (WS) contribute to the total debris volume.

It should be noted that prior research, including the work conducted by the authors, has revealed that existing models for estimating debris tend to yield significant errors in estimating debris volume [55,56]. In contrast, the proposed workflow and debris estimation approach demonstrate higher reliability in cases of complete building destruction, exhibiting a significantly decreased estimation error in comparison to pre-existing models. Additionally, the proposed human-AI teaming approach can provide an informative estimation of debris volume and composition with minimal errors.

5. Discussion

This study holds broader implications for decision-making involved in disaster response and recovery planning. Accurate debris estimation insights support the selection of effective strategies to reduce debris burdens and optimize resource allocation. Understanding the volume and composition of debris enables decision-makers to design targeted waste management plans, including recycling initiatives, to minimize environmental impacts and public health risks. Moreover, the study's findings may be useful for informing the development of comprehensive and efficient disaster recovery decision-making, enhancing the environmental and economic sustainability of affected communities [72]. By integrating accurate debris estimation into disaster management practices, stakeholders will be able to make well-informed decisions that contribute to effective recovery and long-term sustainability.

The access to field-validated ground-truth debris volume data after the disaster events, e.g., Hurricane Harvey and Hurricane Laura, remains limited. The absence of precise and comprehensive data on debris volume post-disaster poses a challenge in directly validating the proposed workflow's accuracy. While efforts were made to validate the framework using available data sources and crowdsourcing methods, the lack of ground-truth data introduces a degree of uncertainty in the validation of our approach. In future work, collaborating with disaster response agencies and government entities (e.g., FEMA and state disaster management agencies) to create comprehensive databases of debris volume data from past events can assist in refining and validating the proposed workflow.

Besides, while the presented human-AI teaming approach shows promising effectiveness, there are several limitations that need to be acknowledged. Firstly, the study relies on the assumptions and accuracy of the damage assessment results obtained through the human-AI teaming system. Despite efforts made to ensure data integrity and quality control, inherent subjectivity and variability in participant responses may introduce some degree of uncertainty in the debris estimation process. Factors such as variations in participant expertise, subjective judgment, or inconsistencies in the labeling process could impact the reliability of the results. To address this, potential future works include focusing on refining the crowdsourcing survey and process by implementing additional quality control measures and validation techniques. This could involve developing standardized guidelines, conducting inter-rater reliability assessments, and incorporating expert input to validate the assessments.

In recent years, photogrammetry has risen as a useful tool in the field of disaster research. Photogrammetry is a methodology for creating precise 3D models and maps by scrutinizing overlapping images, typically acquired through drones or aerial photography [73,74]. It can potentially hold the promise of enhancing damage assessment and streamlining search and rescue efforts. For example, Jafari et al. [73] harnessed smartphone-based photogrammetry to forecast debris volumes after Hurricane Harvey in 2017. It's also worth mentioning that the authors recently leveraged UAV-based aerial imagery and photogrammetry to model a built environment, suggesting that the exploration of photogrammetry-based debris estimation from aerial imagery is a potential future research direction stemming from this paper [13].

The proposed debris estimation approach and discussions about disaster management are based on a specific disaster event Hurricane Laura. The event has unique characteristics and regional factors that may limit the generalizability of the findings to other types of disasters or geographical regions [75]. Each disaster may result in varying types and quantities of debris, necessitating tailored approaches and considerations. To overcome this limitation, future research should expand the scope of the study to incorporate a wider range of disaster events and geographical regions. Examining different types of disasters, such as earthquakes or wildfires, will provide a more comprehensive understanding of the proposed approach's applicability and effectiveness, considering the specific challenges and variations in debris characteristics that each disaster presents.

Finally, while the workflow offers fast and relatively accurate estimates of debris volume and composition, there are inherent limitations and uncertainties associated with debris estimation in post-disaster scenarios. Factors such as hidden or buried debris, inaccessible areas, or variations in debris collection methods can affect the accuracy of the estimates. To address these challenges, the introduced workflow should be further validated through extensive field studies and real-world implementation. In terms of debris recycling and disaster management, future research can explore practical implementation strategies and assess their feasibility in actual scenarios. Also, finetuning AI models to handle diverse disaster scenarios, enhancing crowdsourcing techniques for better data quality, integrating real-time data, and validating the workflow across different disaster types are crucial steps. Additionally, cost-benefit analyses can help justify widespread adoption, facilitating continued innovation and effectiveness in disaster management.

Collaborating with waste management authorities (e.g., FEMA, TDEM, and DTFL), environmental agencies, and relevant stakeholders can facilitate the development of comprehensive recycling frameworks while evaluating their environmental and economic impacts. The authors anticipate that, by incorporating accurate and informative debris estimation into decision-making processes, stakeholders can make informed choices and optimize resource allocation for effective recovery and long-term sustainability efforts.

6. Conclusion

Disaster events generate substantial debris and waste, necessitating extensive and costly cleanup operations post-event. Effectively managing this debris is critical, but it is hampered by resource limitations, high costs, and the diverse nature of the debris. Accurate estimation of debris volume and composition is essential for optimized resource allocation, efficient waste management, and informed decision-making throughout the disaster response and recovery phases. However, traditional methods of field-based preliminary damage assessment and debris estimation are beset by time constraints, human effort, and accuracy issues.

To address these challenges, this study introduced an innovative human-AI teaming workflow that utilizes sensing technologies, AI techniques, and crowdsourcing to automate and expedite debris estimation. The conducted case study in Lake Charles, LA, served as a testbed for the proposed approach, confirming its effectiveness and reliability. The results show that, by integrating advanced technologies and methodologies into debris estimation, the efficiency and efficacy of post-disaster recovery efforts can be significantly enhanced. This advancement will contribute to the establishment of more resilient and sustainable communities, better prepared to mitigate and recover from climate-related hazards.

While this study has demonstrated promising advancements, its limitations include data quality, real-world ground-truth validation, reliability, and generalizability to other disasters in broader implementation. Also, implementing the proposed workflow may face challenges such as technological barriers and the need for training emergency response teams. Potential solutions or future research directions include developing user-friendly AI tools, providing training programs for responders, and ensuring robust data infrastructure for integration of AI and crowdsourcing technologies. Nonetheless, the findings present an opportunity for decision-makers to devise and implement effective recovery strategies, fostering a proactive and well-informed approach to mitigating the adverse effects of natural hazards. Overall, the proposed human-AI teaming workflow will improve decision-makers in optimizing resource allocation, cost reduction, and overall improvement of disaster response and recovery outcomes.

CRediT authorship contribution statement

Chih-Shen Cheng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Amir Behzadan:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Arash Noshadravan:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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